

Computing Basics

The tools of the trade

10/05/22

The age of computing



1964: CDC 6600

“First” supercomputer

Capable of 1 megaFLOP (1 million)

FLOPs= operations (+,-,X) per second



2014: iPhone 6

~1000 megaFLOPS

2022: iPhone 13

~768,000 megaFLOPS

How large are machines now?

■
1 Pf machine
NEU "Discovery"
~30000 cores



iPhone

Area of each signifies
the performance

~200 Pflop machines (2015)

Titan

Sequoia

JUGEEN

MilkyWay-2

Sunway TiahuLight

The current leader (2022)



Frontier Supercomputer at ORNL

More than 600,000 CPU cores and 37,000 GPUs

Fiber optic connections between the cores

Capable of 1,000,000,000,000,000,000 calculations per second

~10,000,000,000,000 times more powerful than the CDC 6600!!!!

Weights approximately 200,000 kg!

Data Storage Yesterday (1956): 5 megabyte IBM data storage drive = 5,000,000 bytes



[viralfhq](#)

A five-megabyte hard drive is shipped by IBM, 1956.

Modern (small) Data Storage



~\$20

128 Gb thumb drive

~25,000 times more storage than the
IBM storage drive

Data Storage Today, 1.5 petabytes data storage system= 1,500,000,000,000,000 bytes

Sony whips up a box to store 1.5 petabytes of data on 10,000 Blu-ray discs

JORDAN NOVET MARCH 10, 2015 2:33 PM

TAGS: BLU-RAY, COLD STORAGE, FACEBOOK, OPEN COMPUTE PROJECT, OPEN COMPUTE SUMMIT, SONY



Above: Sony's new prototype for storing 1.5 PB of data on Blu-ray discs at the Open Compute Summit in San Jose, Calif., on March 10.

This system can store about 300,000,000 times the amount of data that a state-of-the-art IBM (5 megabyte) drive could store in 1956 – see earlier slide

What does this mean?

First teraflop machine appeared in 1996 (ASCI “Red”)

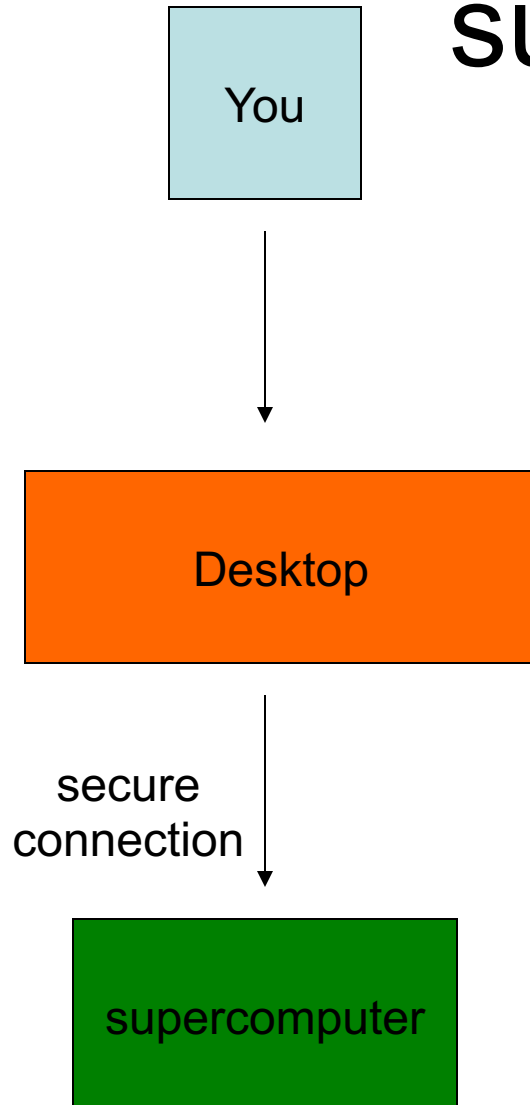
A single GPU can exceed 1 teraflop (~\$2000)

The supercomputing of the early 90s is trivial today

The cost of computation is small enough that anyone can afford to incorporate computational components into their research

**Computers are only tools, we need to ask
the right questions!!!**

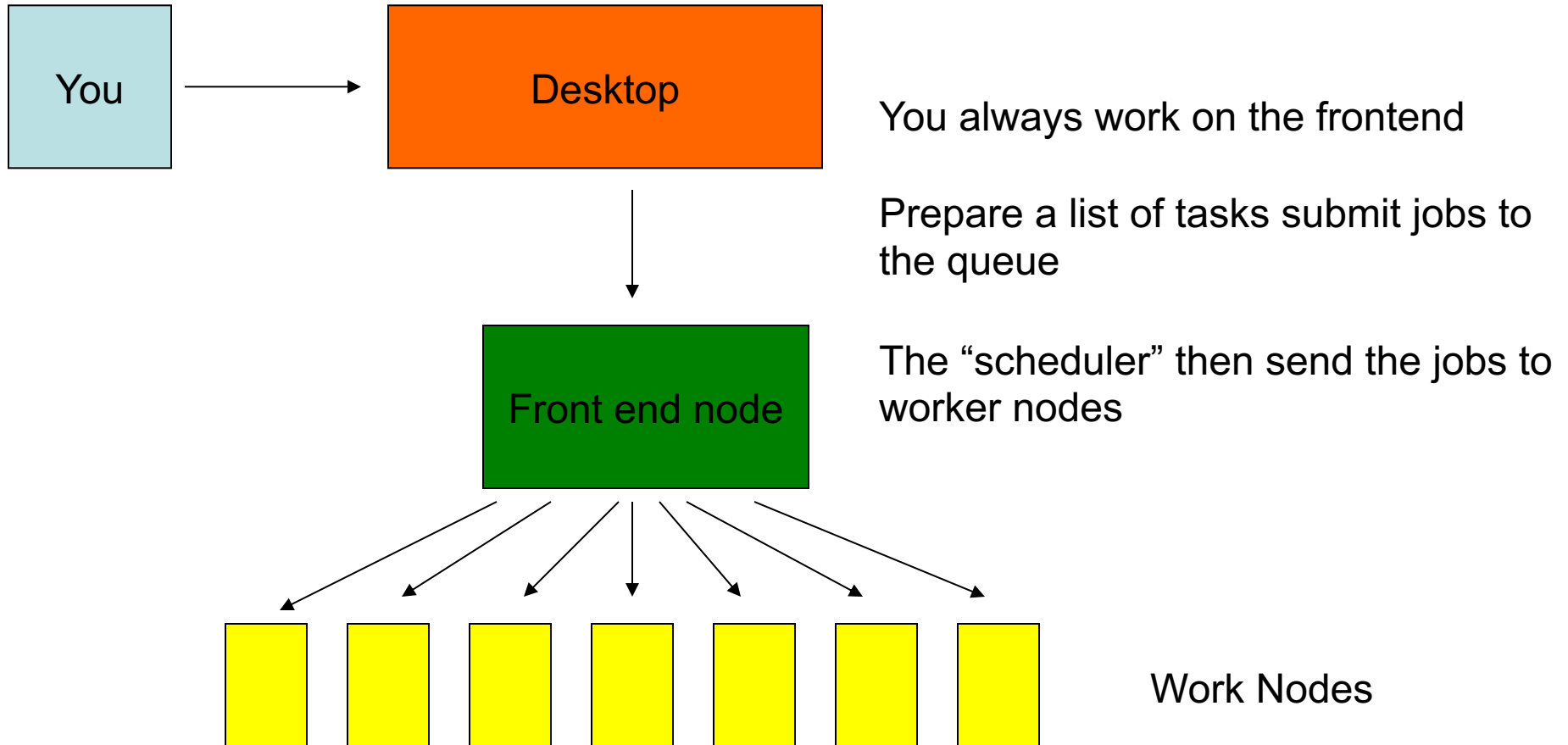
How do you use a supercomputer?



```
pcw2@bgq-fn:~  
Problems and/or questions should be submitted via the problem tracking  
system form: http://helpdesk.rice.edu  
Enter remote host: bgq  
connecting to bgq  
pcw2@bgq's password:   
Last login: Mon Mar 16 14:16:13 2015 from   
-----  
RICE UNIVERSITY BLUEGENE/Q  
=====
```

The screenshot shows a terminal window with a title bar that reads 'pcw2@bgq-fn:~'. The terminal output includes instructions on how to submit problems, prompts for a remote host and password, and a last login message. It then displays a message from RICE UNIVERSITY BLUEGENE/Q stating that unauthorized access is prohibited and providing a URL for questions and problem reports. Finally, it welcomes the user to Bluegene/Q and provides instructions on using the 'module' command. The prompt '[pcw2@bgq-fn ~]\$' is visible at the bottom.

Where is the work done?



The point?

- Using a terminal is the only real option for using high performance computing resources
- We will use a terminal to work locally on your machine, but the ideas can be used to access larger resources in the future

Using a Terminal

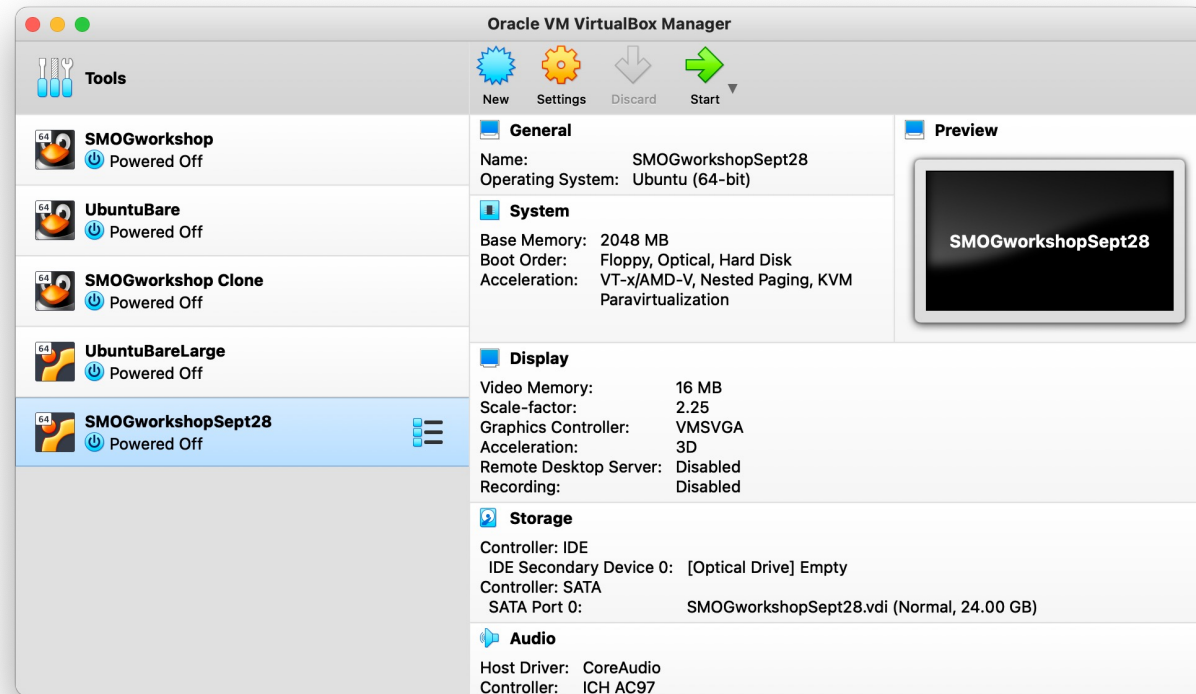
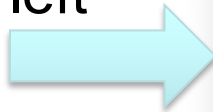
- Today: We will work locally using the terminal
- The terminal is based on **commands**. That is, you can't click on things!
- Here, we are going to use the terminal inside of a virtual machine

Virtual Machines

- A VM is a piece of software that allows you to install a second operating system and use it as if it were an actual computer
- For example:
 - You may have a Windows machine
 - You can install VM software
 - When in the VM, you can install a Linux operating system
 - Then, use the Linux-based VM, even though you are on a Windows machine

Starting the Linux Virtual Machine

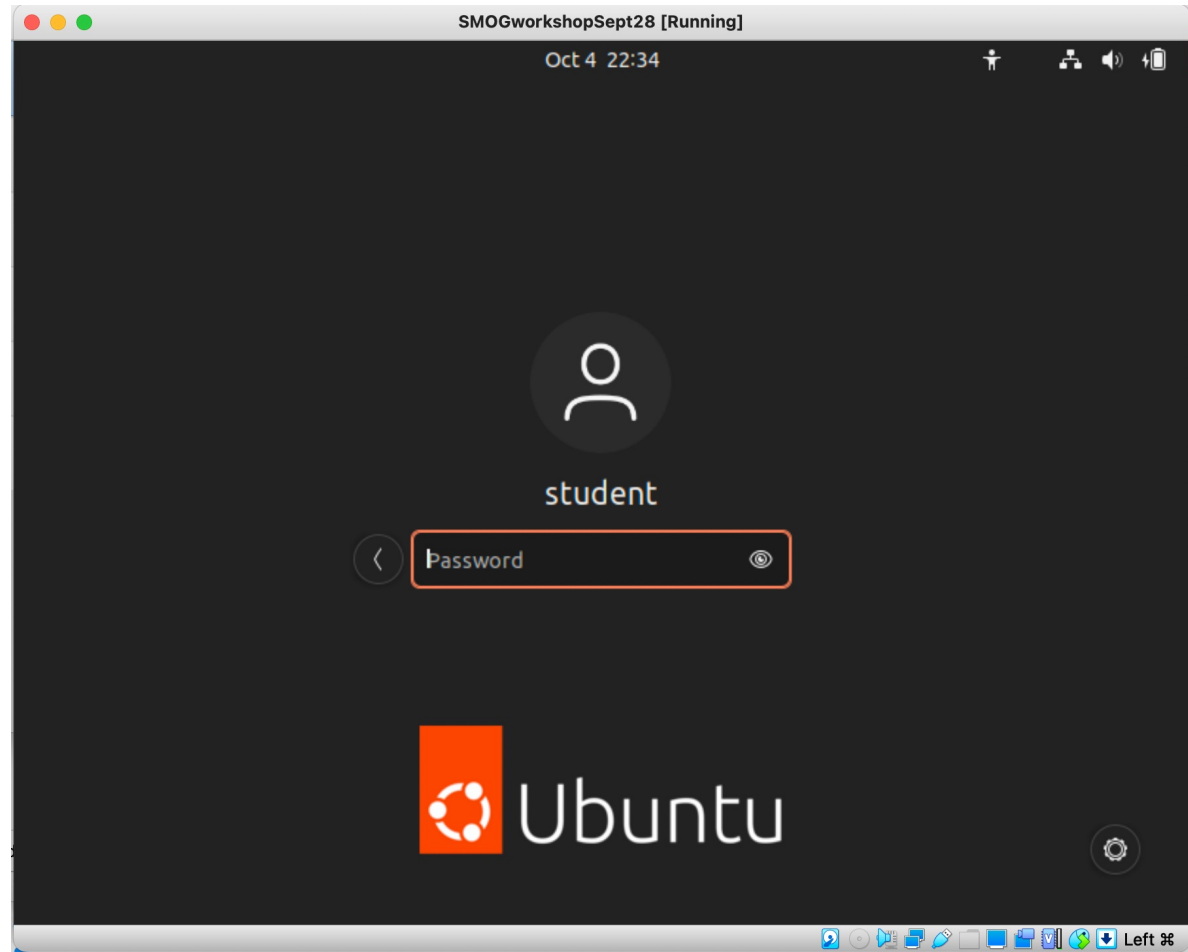
- Start VirtualBox
- If you already imported the image, then simply click on the VM image on the left



If the VM launches, you will see an Ubuntu based OS and a prompt to login

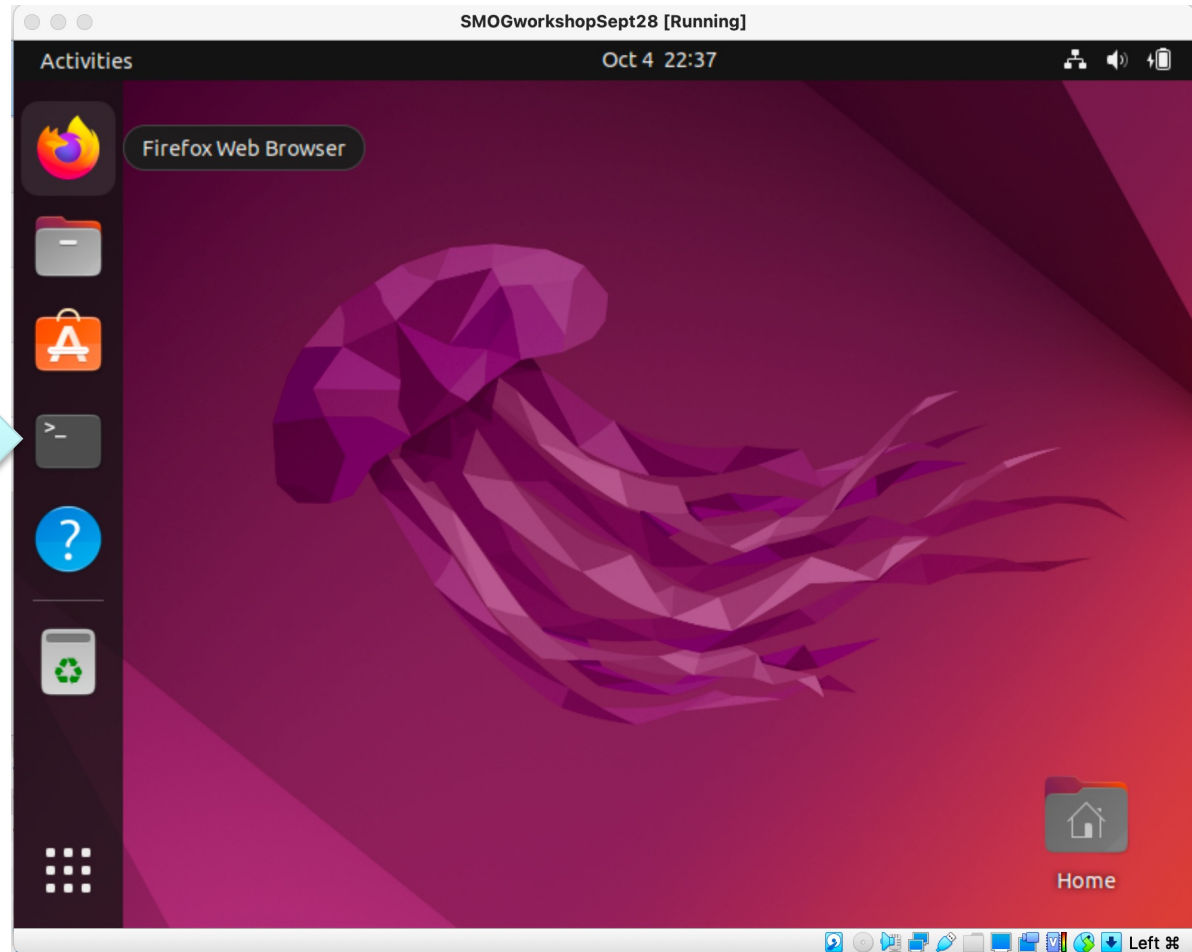
Login with
Username: student
Password: smog

You may be automatically logged in. That is fine. If you get logged out, you can use the credentials above.

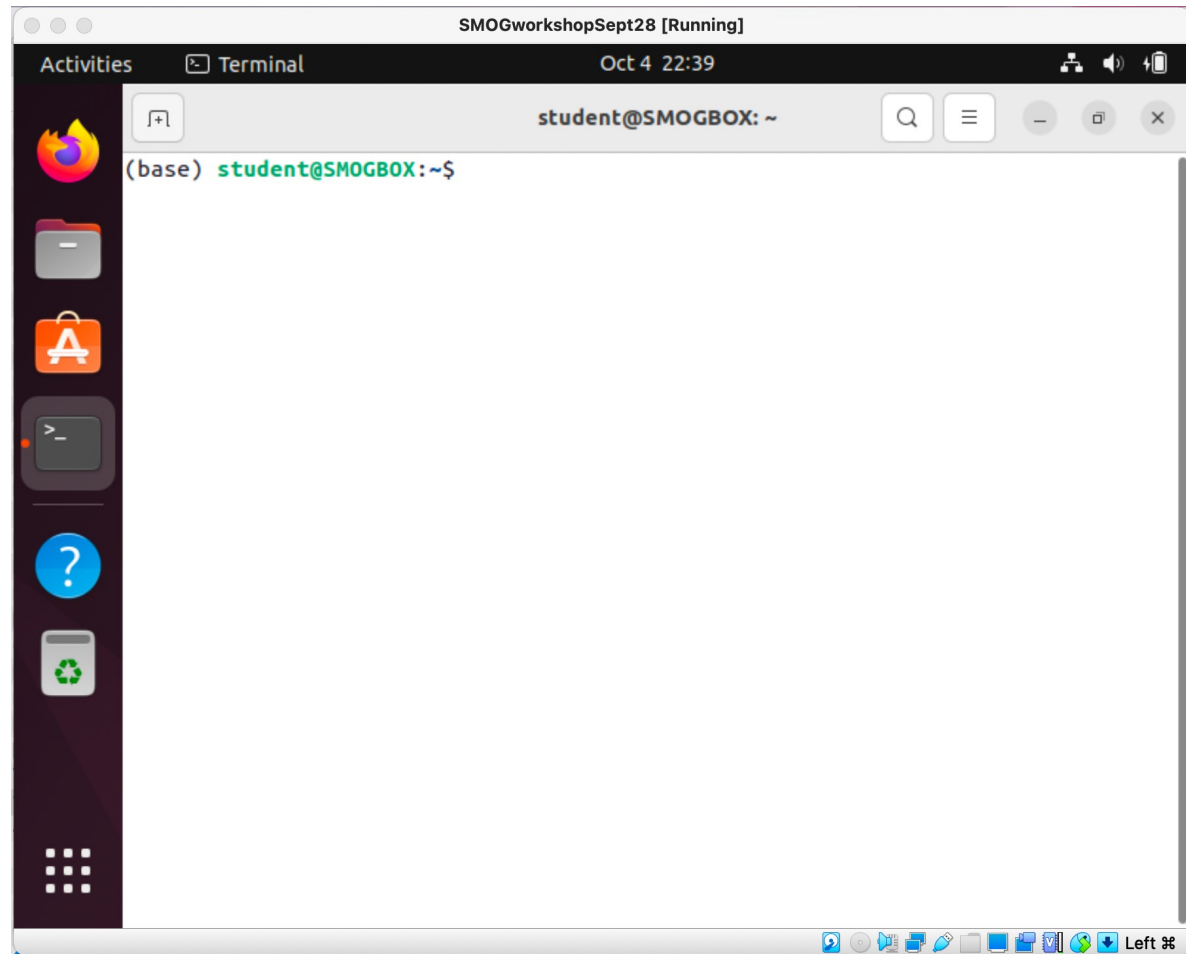


Once logged in, you are in Linux.

Open the terminal by clicking on the gray box



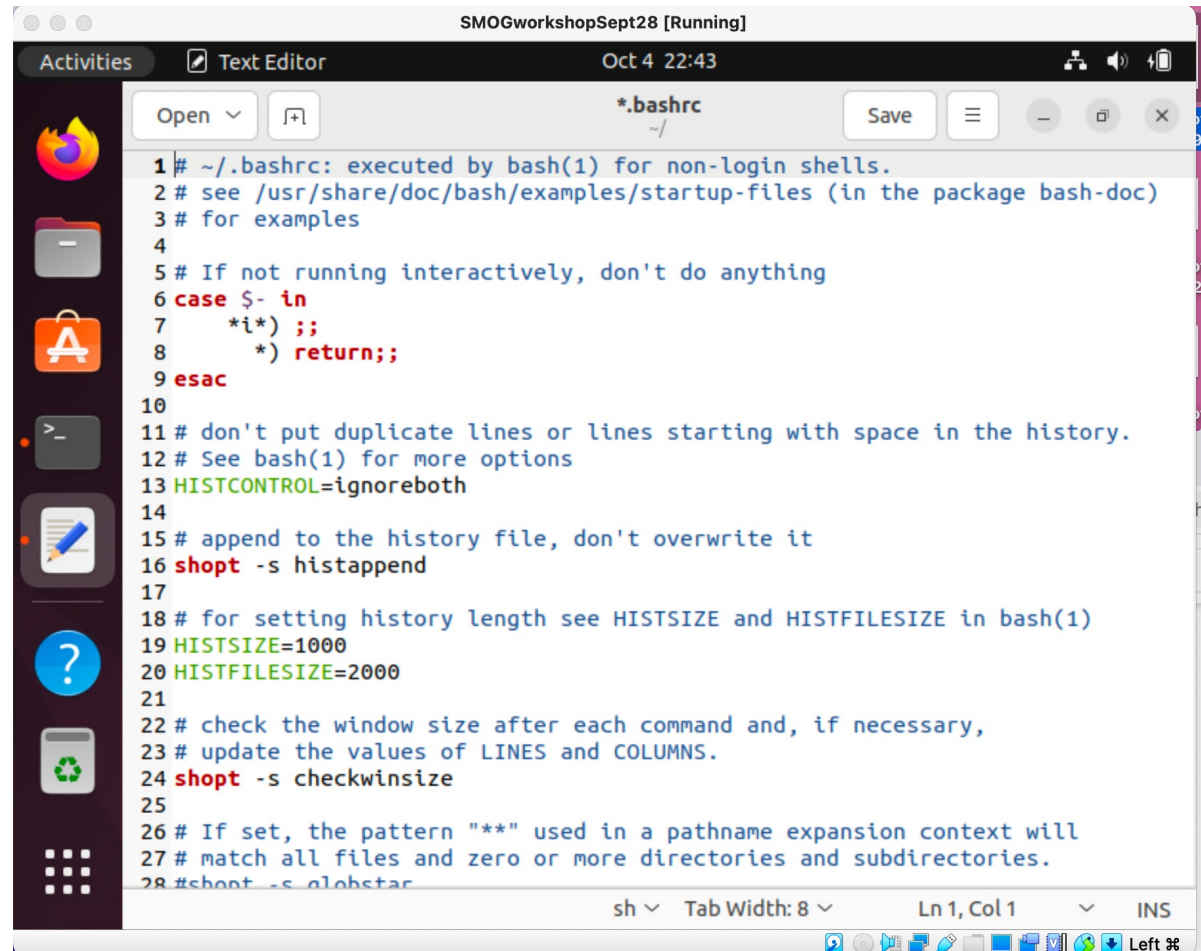
You are now in the terminal



Fixing my mistake

- Due to an oversight, we will need to help configure the working environment, so that all programs will work.
- On the command line, type:
`gedit .bashrc`
The hit enter/return.

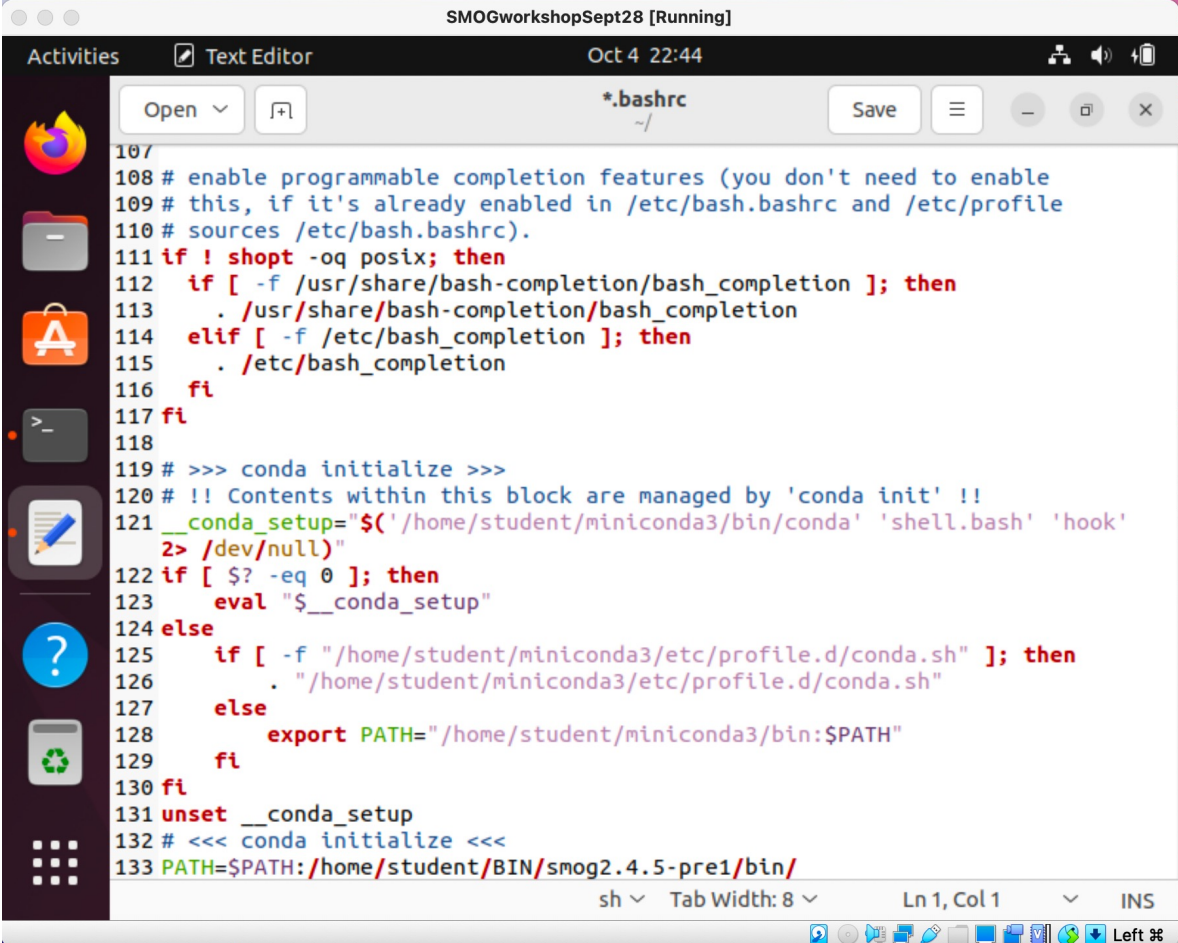
Looking inside the .bashrc file



The screenshot shows a Linux desktop environment with a dark theme. The top panel displays the window title 'SMOGworkshopSept28 [Running]' and the date 'Oct 4 22:43'. The left sidebar contains icons for the Dash, Home, Applications, and Dash to Dock. The main window is a 'Text Editor' showing the contents of the '~/.bashrc' file. The file contains various shell configuration lines, including comments, a case statement for interactive shells, and settings for history and window size. The status bar at the bottom indicates the shell is 'sh', tab width is 8, and the cursor is at line 1, column 1.

```
1 # ~/.bashrc: executed by bash(1) for non-login shells.
2 # see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
3 # for examples
4
5 # If not running interactively, don't do anything
6 case $- in
7     *) ;;
8     *) return;;
9 esac
10
11 # don't put duplicate lines or lines starting with space in the history.
12 # See bash(1) for more options
13 HISTCONTROL=ignoreboth
14
15 # append to the history file, don't overwrite it
16 shopt -s histappend
17
18 # for setting history length see HISTSIZE and HISTFILESIZE in bash(1)
19 HISTSIZE=1000
20 HISTFILESIZE=2000
21
22 # check the window size after each command and, if necessary,
23 # update the values of LINES and COLUMNS.
24 shopt -s checkwinsize
25
26 # If set, the pattern "*" used in a pathname expansion context will
27 # match all files and zero or more directories and subdirectories.
28 shopt -s globstar
```

Editing the .bashrc file



```
107
108 # enable programmable completion features (you don't need to enable
109 # this, if it's already enabled in /etc/bash.bashrc and /etc/profile
110 # sources /etc/bash.bashrc).
111 if ! shopt -oq posix; then
112     if [ -f /usr/share/bash-completion/bash_completion ]; then
113         . /usr/share/bash-completion/bash_completion
114     elif [ -f /etc/bash_completion ]; then
115         . /etc/bash_completion
116     fi
117 fi
118
119 # >>> conda initialize >>>
120 # !! Contents within this block are managed by 'conda init' !!
121 __conda_setup="$(('/home/student/miniconda3/bin/conda' 'shell.bash' 'hook'
122 2> /dev/null)"
123 if [ $? -eq 0 ]; then
124     eval "$__conda_setup"
125 else
126     if [ -f "/home/student/miniconda3/etc/profile.d/conda.sh" ]; then
127         . "/home/student/miniconda3/etc/profile.d/conda.sh"
128     else
129         export PATH="/home/student/miniconda3/bin:$PATH"
130     fi
131 unset __conda_setup
132 # <<< conda initialize <<<
133 PATH=$PATH:/home/student/BIN/smog2.4.5-pre1/bin/
```

Go to last line and add:

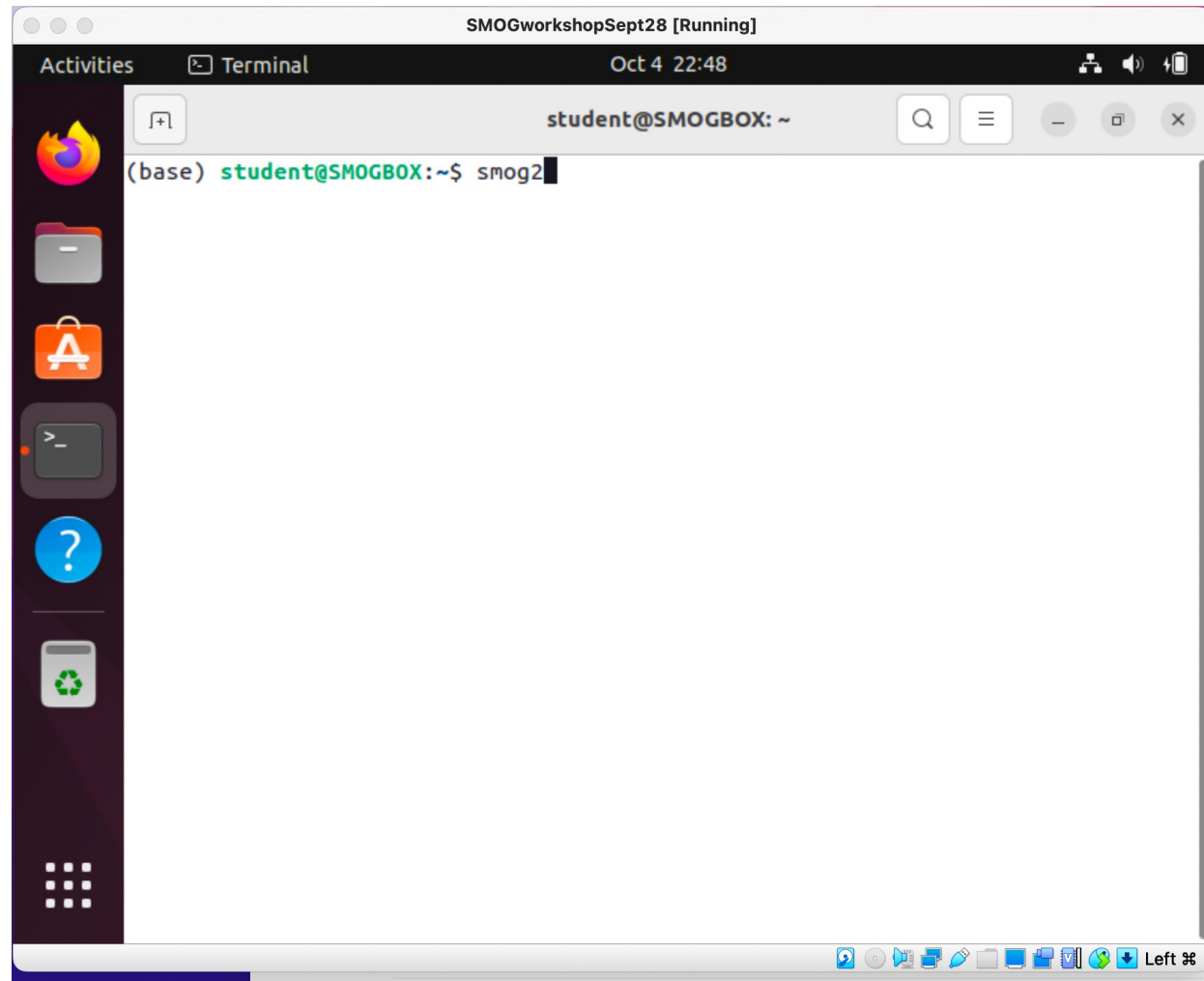
PATH:\$PATH:/home/student/BIN/smog2.4.5-pre1/bin

Hit “save”, close the file and close the Terminal

Testing your edit...

Once the Terminal is closed, reopen it and type “smog2”, then hit enter.

If everything worked, then you will see a bunch of text on the screen.



Trying other useful Linux commands

- Linux is command based
 - i.e. You enter text (commands) and hit “enter” key if you want something to happen
- Common Commands (follow along)
 - Make a directory: `mkdir TEMPDIR`
 - Change directory (going into TEMPDIR) “`cd TEMPDIR`”
 - List contents of directory: `ls`
 - You should see nothing
 - View a file `Stuff.txt`: “`gedit Stuff.txt &`”
 - Enter text, save and close
 - Remove file `Stuff.txt`: “`rm Stuff.txt`”
 - Get out of directory: “`cd ..`”
 - Remove directory: “`rmdir TEMPDIR`”
 - Show where you are: “`pwd`”

Programs we will use tomorrow

All of these tools are already installed in the VM.

- VMD: Visual molecular dynamics (available in the VM)
 - To start session, type “vmd” in the Terminal
- SMOG 2
 - This will be used to generate models from structure
- OpenSMOG/OpenMM
 - This will be used to perform simulations